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The composite-disturbance-observer based stochastic model predictive control for spacecrafts under multi-source disturbances

Yang Xu[a], Yuan Yuan[a,*], Dalin Zhou[b]

[a] School of Astronautics, Northwestern Polytechnical University, 710072, China
[b] School of Computing, University of Portsmouth, Portsmouth PO1 3HE, U.K

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Abstract

In this paper, the attitude control problem of the spacecraft system under input/state constraints and multi-source disturbances is investigated. A novel estimation method, composite-disturbance-observer (CDO), is proposed to provide an estimate for both modeled and unmodeled disturbances in an online manner. Based on the estimates provided by the CDO, the composite stochastic model predictive control (C-SMPC) scheme is designed for attitude control. The recursive feasibility of the C-SMPC method is guaranteed by reformulating the state and input constraints. Furthermore, the sufficient conditions are established to guarantee the stability of the overall closed-loop system. Finally, the simulation on the attitude control of the spacecraft is conducted to verify the effectiveness of the proposed method.
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1. Introduction

Recent years have witnessed a constant research attention on the attitude control of spacecrafts due mainly to its significance in the critical orbital missions [1]. To accomplish these missions, spacecrafts may need to rotate along a relatively large-angle amplitude trajectory, which can be described by the nonlinear differential equations [2]. Additionally, in the attitude control problem, the uncertainty of spacecraft mass, the onboard payload motion, the rotation

* Corresponding author.
E-mail address: snowkey@aliyun.com (Y. Yuan).

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of solar arrays and fuel consumption have to be simultaneously taken into consideration [3]. These two factors make the attitude control of spacecrafts extremely difficult and motivate our current investigation.

Unlike the success in an ideal environment, a practical control system is not guaranteed to function in real disturbance rejection problem and remains critical to be investigated. The problem has been significantly addressed with an improved controller design. Among all the feasible candidates, the disturbance observer (DO) [4–8] driven solutions have been widely adopted to utilize its estimation for the disturbance. If the disturbance can be described by an exogenous system (modeled disturbance), the DO could make use of the information of the exogenous system and it is possible that the error dynamics of the DO system is asymptotically stable [9]. In [10], a disturbance observer has been proposed to deal with the disturbances composed of the harmonic signals and signals with bounded H_2-norm. Besides, in [11], a disturbance observer has been constructed to estimate the disturbance with partially-known information under the white noise. On the other hand, if the disturbance can not be expressed by an exogenous system (unmodeled disturbance), the unmodeled disturbance can be treated as a slowly time-varying factor [12] yet not to be accurately estimated by the DO. In this case, the estimation error of the DO system can only be guaranteed to be bounded. To deal with the unmodeled disturbances, some other DO have been proposed. In [13], an adaptive extended state observer (ESO) has been designed for nonlinear disturbed systems to get better flexibility and performance than the traditional linear ESO. In [14], an ESO has been proposed to estimate both state, and total disturbance which has included the internal uncertain nonlinear part and the external uncertain stochastic disturbance. Besides, in [15], a learning-based DO has been augmented by a learning scheme, which has been motivated by iterative learning control, to further enhance the estimation and suppression of the disturbance when it has had repetitive components. Furthermore, in [16], a DO has been developed to provide efficient learning of the compounded disturbance which has included the effect of time-varying disturbance, fuzzy approximation error, and unknown dead zone. In [17], a disturbance-observer-based control in combination with a neural network scheme and back-stepping method has been developed to achieve a composite anti-disturbance controller design that provides guaranteed performance. Despite the factual situation that engineering systems are subject both modeled and unmodeled disturbances simultaneously in practice, only a limited number of research have considered the multi-source disturbances [4,5,9]. Instead, most existing literature on the DO has been dedicated to solving problems under a unimodal (either modeled or unmodeled) disturbance. In this paper, a novel composite-disturbance-observer (CDO) is proposed to underpin the significance of incorporating multi-source disturbances in the control problem by providing simultaneous estimate both modeled and unmodeled disturbances. The novel CDO forms a main contribution of this paper and underlies the rest research.

Let alone the multi-source nature of disturbances in practice, their stochastic property remains another challenging issue to be addressed. As pointed out in [18], by considering the state and input constraints in a probabilistic manner, the frequencies of the behaviors leading to the equipment fatigue could be significantly reduced. As a result, the life of the facilities can be en-longed. It has been well recognized that the subsystems/facilities of the spacecraft are not easy to be replaced (or even repaired) [19]. As such, it is of significance to investigate the control problem of the spacecraft subject to the probabilistic state and input constraints. Out of a number of candidates, stochastic model predictive control (SMPC) method is recognized as effective to accommodate stochastic disturbances. Different from the SMPC, the robust MPC

has also been used to deal with uncertainties [20–23]. In [20], the robust distributed MPC has been presented for a class of linear systems subject to structured time-varying uncertainties. In [21], the robust distributed MPC has been proposed for a class of linear discrete-time systems with unknown parameters. The self-triggered robust MPC has been designed for nonholonomic vehicle with coupled input constraint and bounded disturbances [22]. Compared with traditional robust MPC, the SMPC is capable of handling the stochastic noise by using the distribution information [24]. In [25], the authors have proposed a Lyapunov-based model predictive controller for nonlinear systems subject to stochastic uncertainty, and the stabilization in probability is established. In [26], the stochastic Lyapunov-based economic model predictive control has been proposed for a broad class of stochastic nonlinear systems with input constraints. Despite the effectiveness of SMPC and the inherent disturbances in the attitude control of spacecrafts, there is no bridging research in literature. This paper is presented as the first attempt to fill the gap.

Summarizing the above discussions, in this paper, we are endeavored to investigate the attitude control problem of the spacecraft subject to probabilistic state and input constraints as well as the multi-source disturbances including modeled/unmodeled disturbance and stochastic noise. To be more specific, The composite SMPC (C-SMPC) is designed which constitutes of the SMPC and the estimate provided by the CDO. The main contributions of our paper are highlighted as follows: *1) the CDO can not only get the estimate of the disturbance generated by the exogenous system, but also enables to get the estimate of unknown nonlinearities; 2) the noise-to-state stability in probability of the CDO is guaranteed by the proposed sufficient conditions; and 3) the recursive feasibility and mean square stability of the closed-loop system are guaranteed under the proposed C-SMPC scheme.*

Notation: In this paper, $\mathbb{Z}_{\geq 0}$ represent the natural number set. $\mathbb{Z}^+$ represents the positive integer set. $\mathbb{Z}[a, b]$ with $a, b \in \mathbb{Z}_{\geq 0}$ represents the positive integer set $\{a, a+1, \cdots, b\}$. $\mathbb{R}^n$ denotes the n-dimensional Euclidean space. For any matrix A, $A \in \mathbb{R}^{n\times m}$ denotes the A is a $n \times m$-dimensional matrix. A^T denotes the transpose of matrix A. Denote $\|x\|$ as the Euclidean norm of x and $\|x\|_P \triangleq \sqrt{x^T P x}$ as the P-weighted norm of x. $\mathbf{0}_{m\times n}$ represents the $m \times n$ dimensional matrix full of 0. $\mathbf{I}_{m\times n}$ represents the $m \times n$ dimensional matrix full of 1. 1^+ represents the right limit of 1. $*$ represents the symmetric part in the matrix.

2. Problem formulation and preliminaries

2.1. System model

Consider the attitude dynamics of the spacecraft in the body frame [39]:

$$J_{s/c}\dot{\omega} - (J_{s/c}\omega) \times \omega = \mathbf{u} + d_{\text{ext}} \tag{1}$$

where $\omega \in \mathbb{R}^3$ is the angular velocity; $J_{s/c} \in \mathbb{R}^3$ is the rotational inertial of the spacecraft; $\mathbf{u} \in \mathbb{R}^3$ and $d_{\text{ext}} \in \mathbb{R}^3$ are the control input and external disturbance, respectively. In actual practice, the change of $J_{s/c}$ is negligible, i.e., $\dot{J}_{s/c} = 0$.

In this paper, the Modified Rodriguez Parameters (MRPs) are used to describe the orientation of the rigid body relative to the inertial frame [27]. Then, we have

$$\dot{\boldsymbol{q}} = \boldsymbol{Z}(\boldsymbol{q})\omega \tag{2}$$

where

$$\boldsymbol{Z}(\boldsymbol{q}) = \frac{1}{2}\left[\mathbf{I}_{3\times 3}\left(\frac{1-\boldsymbol{q}^T\boldsymbol{q}}{2}\right) + \boldsymbol{q}\boldsymbol{q}^T + \boldsymbol{S}(\boldsymbol{q})\right]$$

with $\boldsymbol{q} = [q_1\ q_2\ q_3]^T$ and the skew-symmetric matrix function

$$\boldsymbol{S}(\boldsymbol{q}) = \begin{bmatrix} 0 & -q_3 & q_2 \\ q_3 & 0 & -q_1 \\ -q_2 & q_1 & 0 \end{bmatrix}$$

According to Chung et al. [28], based on (1) and (2), the following Lagrangian formulation of attitude dynamics is obtained in terms of the MRPs:

$$\ddot{\boldsymbol{q}} = \boldsymbol{M}^{-1}(\boldsymbol{q})\left[-\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \tau_{\text{ext}}\right] + \boldsymbol{M}^{-1}(\boldsymbol{q})\tau_u \tag{3}$$

with

$$\tau_u = \boldsymbol{Z}^{-T}(\boldsymbol{q})\mathbf{u},\ \ \tau_{\text{ext}} = \boldsymbol{Z}^{-T}(\boldsymbol{q})d_{\text{ext}},\ \ \boldsymbol{M}(\boldsymbol{q}) = \boldsymbol{Z}^{-T}(\boldsymbol{q})J_{s/c}\boldsymbol{Z}^{-1}(\boldsymbol{q})$$
$$\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}}) = -\boldsymbol{M}(\boldsymbol{q})\dot{\boldsymbol{Z}}(\boldsymbol{q})\boldsymbol{Z}^{-1}(\boldsymbol{q}) - \boldsymbol{Z}^{-T}(\boldsymbol{q})\boldsymbol{S}(J_{s/c}\boldsymbol{Z}^{-1}(\boldsymbol{q})\dot{\boldsymbol{q}})\boldsymbol{Z}^{-1}(\boldsymbol{q})$$

To begin with, we consider the case that d_{ext} is composed of three types of disturbances: the deterministic modeled disturbance $d_o(t)$, stochastic noise $d_s(t)$ and a norm bounded disturbance $d_b(t)$, i.e., $d_{\text{ext}} = d_o(t) + d_s(t) + d_b(t)$. We assume that $d_o(t)$ is generated by the following exogenous system:

$$\dot{w}(t) = \tilde{W}w(t) + \tilde{H}\eta(t),\ \ d_o(t) = \tilde{V}w(t) \tag{4}$$

where $w(t) \in \mathbb{R}^3$ is the auxiliary state of the exogenous system. According to Guo and Chen [4], d_o normally represents harmonic (or constant) disturbance and hence we suppose $\|w(t)\|_2 \le \bar{w}$. The matrices $\tilde{W}$, $\tilde{H}$ and $\tilde{V}$ are known. $\eta(t) \in \mathbb{R}^6$ represents the uncertainties in the exogenous system with $\eta(t) \in l_2[0,\infty)$ and we suppose $\|\eta(t)\|_2 \le \bar{\eta}$. Additionally, we take into consideration of the stochastic noise $d_s(t)$ satisfying

$$\begin{aligned} &\mathbf{Pr}\{d_{s,i}(t) \le \eta_{s,i}\} = F_{s,i}(\eta_{s,i}) \\ &F_{s,i}(\eta_{s,i}) = \begin{cases} 1, & \text{for}\,\eta_{s,i} \ge \alpha_{s,i} \\ 0, & \text{for}\,\eta_{s,i} < -\alpha_{s,i} \end{cases} \end{aligned} \tag{5}$$

where $\alpha_{s,i} > 0$, $i \in \{1,2,3\}$. Denote $d_{s,i}(t)$ as the ith component of $d_s(t)$. Note that $d_{s,1}(t)$, $d_{s,2}(t)$ and $d_{s,3}(t)$ are mutually independent. The distribution function $F_{s,i}(\cdot)$ is continuous on the right with only a finite number of discontinuities. The distribution rule (5) suggests that $d_{s,i}(t)$ lie in the orthotope, namely, $d_{s,i}(t) \in \Theta_{s,i} = \left\{d_{s,i} : |d_{s,i}| \le \bar{d}_{s,i}\right\}$. In other words, we have $\|d_s(t)\|_2 = \sqrt{\sum_{i=1}^3 |d_{s,i}|^2} \le \sqrt{\sum_{i=1}^3 \bar{d}_{s,i}} = \bar{d}_s$. Denote the expectation and variance of $d_s(t)$ as $\tilde{d}_s = [\tilde{d}_{s,1}\ \tilde{d}_{s,2}\ \tilde{d}_{s,3}]^T$ and $\breve{d}_s = [\breve{d}_{s,1}\ \breve{d}_{s,2}\ \breve{d}_{s,3}]$, respectively. $d_b(t)$ satisfies $\|d_b(t)\|_2 \le \bar{d}_b$. Denote $x_1 = q_1$, $x_2 = q_2$, $x_3 = q_3$, $x_4 = \dot{q}_1$, $x_5 = \dot{q}_2$, $x_6 = \dot{q}_3$, $d_f = \boldsymbol{M}^{-1}(\boldsymbol{q})\left[-\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \tau_{\text{ext}}\right] - d_o - d_s$ and $u = \boldsymbol{Z}(\boldsymbol{q})J_{s/c}^{-1}\mathbf{u}$. It is worth pointing out that $d_f(t)$ can be regarded as the unmodeled disturbance. According to Properties 1–4 in [29], $\boldsymbol{M}^{-1}(\boldsymbol{q})$, $\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}}$, τ_{ext}, d_o and d_s are all bounded. Therefore, $d_f(t)$ is bounded with $\|d_f(t)\|_2 \le \bar{d}_f$. System (3) can be rewritten compactly as:

$$\begin{cases} \dot{x}(t) &= \tilde{A}x(t) + \tilde{B}u(t) + \tilde{G}d_f(t) + \tilde{G}d_o(t) + \tilde{G}d_s(t) \\ y(t) &= \tilde{C}x(t) \end{cases} \tag{6}$$

where

$$\tilde{A} = \begin{bmatrix} \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} \\ \mathbf{0}_{3\times3} & \mathbf{0}_{3\times3} \end{bmatrix}, \ \tilde{B} = \tilde{G} = \begin{bmatrix} \mathbf{0}_{3\times3} \\ \mathbf{I}_{3\times3} \end{bmatrix}$$
$$\tilde{C} = \begin{bmatrix} \mathbf{I}_{3\times3} & \mathbf{0}_{3\times3} \end{bmatrix}$$

By discrediting (4), we have

$$w(k+1) = Ww(k) + H\eta(k), \ d_o(k) = Vw(k)$$

where $W = e^{\tilde{W}T}$, $H = \int_0^T \tilde{H}e^{\tilde{W}\tau}\mathrm{d}\tau$, $V = \tilde{V}$ with T being the sampling interval. According to Chang [30], if the sampling period T is small enough, we have $\|d_f(k+1) - d_f(k)\|_2 \leq \|O(T^2)\|_2 \leq \delta_f$ where δ_f is a constant. Up to now, we could provide the discrete-time model for the spacecraft as follows:

$$\begin{cases} x(k+1) & = Ax(k) + Bu(k) + G(d_f(k) + d_o(k) + d_s(k) + O(T^2)) \\ y(k) & = Cx(k) \end{cases} \tag{7}$$

where $A = e^{\tilde{A}T}$, $B = \int_0^T \tilde{B}e^{\tilde{A}\tau}\mathrm{d}\tau$, $G = \int_0^T \tilde{G}e^{\tilde{A}\tau}\mathrm{d}\tau$, $C = \tilde{C}$. Considering the probabilistic constraints, we have

$$\mathbf{Pr}\{b_x^T x(k) \geq h_x\} \leq l_x \tag{8}$$

$$\mathbf{Pr}\{b_u^T u(k) \geq h_u\} \leq l_u \tag{9}$$

where b_x, b_u, h_x and h_u are given constant vectors with appropriate dimensions; l_x and l_u represent probabilities with $l_x \in [0, 1]$ and $l_u \in [0, 1]$, respectively.

Up to now, we are ready to present the design objectives of this paper: 1) design the CDO which could provide an estimate for both $d_o(k)$ and $d_f(k)$; 2) design the SMPC strategy to deal with $d_s(k)$ and probabilistic constraints; 3) design the composite SMPC strategy which includes SMPC and the estimates provided by the CDO; and 4) establish sufficient conditions to guarantee the recursive feasibility and the stability of the overall closed-loop system.

Remark 1. The analysis of the complicated characteristics of the lumped disturbance is shown as: 1) the nonlinearities in the paper contains the unexpected nonlinear signals, modeled disturbances and stochastic disturbances, which is difficult to deal with by only using the disturbance observer; 2) though the extended state observer (ESO) can deal with these complex disturbances, the ESO cannot obtain the accurate estimations of them when the information of the disturbances is limited. Therefore, in this paper, the nonlinearities is divided into three parts: the unexpected nonlinear signals, modeled disturbances and stochastic disturbances; 3) by this step, the nonlinear system is converted to the linear system, which simplifies the system form and classifies the disturbances finely.

Remark 2. The modeled disturbance $d_o(t)$ represents the elastic disturbance introduced by the flexible devices on spacecraft such as spacecraft antenna and solar remake [31]. The norm bounded disturbance $d_b(t)$ represents the Gravity-gradient torque, solar radiation, Earth-magnetic torque (all could be assumed bounded), and aerodynamic torque [32]. The stochastic noise $d_s(t)$ represents the actuator noise of the spacecraft [33].

3. The design of the CDO

In this part, we aim at designing the CDO so as to estimate the modeled disturbance $d_o(k)$ and unmodeled disturbance $d_f(k)$. In the CDO, to estimate $d_o(k)$, we have

$$\begin{cases} \widehat{w}(k) = & v_w(k) + L_w x(k), \ \hat{d}_o(k) = V\widehat{w}(k) \\ v_w(k+1) = & (W - L_w GV)(v_w(k) + L_w x(k)) \\ & -L_w(Ax(k) + Bu(k)) - L_w G\hat{d}_f(k) \end{cases}$$

Meanwhile, to estimate $d_f(k)$, we have

$$\begin{cases} \hat{d}_f(k) = & v_f(k) + L_f x(k) \\ v_f(k+1) = & (\mathbf{I}_{3\times 3} - L_f G)(v_f(k) + L_f x(k)) \\ & -L_f(Ax(k) + Bu(k)) - L_f GV\widehat{w}(k) \end{cases}$$

where L_w and L_f are observer parameters to be designed; $v_w(k)$ and $v_f(k)$ are auxiliary variables. Denoting $e_w(k) = w(k) - \hat{w}(k)$ and $e_f(k) = d_f(k) - \hat{d}_f(k)$, we have

$$e_{wf}(k+1) = \Omega e_{wf}(k) + \tilde{\eta}_1(k) \tag{10}$$

where $e_{wf}(k) = [e_w^T(k) \ e_f^T(k)]^T$ and

$$\Omega = \begin{bmatrix} W - L_w GV & -L_w G \\ -L_f GV & \mathbf{I}_{3\times 3} - L_f G \end{bmatrix}$$
$$\tilde{\eta}_1(k) = \begin{bmatrix} -L_w G d_s(k) - L_w GO(T^2) + H\eta(k) \\ -L_f G d_s(k) + (\mathbf{I}_{3\times 3} - L_f G)O(T^2) \end{bmatrix}$$

Remark 3. In the CDO, the modeled disturbance $d_o(k)$ is estimated by using information from the exogenious system. On the other hand, the unmodeled disturbance $d_f(k)$ is estimated by only requiring that $\|d_f(k+1) - d_f(k)\|_2 \le \delta_f$. The structure of the CDO suggests that it has the following merits: 1) The CDO is more suitable for the multi-source disturbance including both d_o and d_f. 2) In comparison with traditional DO dedicated to unmodeled disturbance, the CDO makes full use of the information of the disturbance. 3) In comparison with the DO dedicated to modeled disturbance, the CDO could handle a more general class of disturbances which are not necessarily described by the exogenious system. 4) The CDO could boil down to the traditional modeled (or unmodeled) DO if only one type of disturbance ($d_o(k)$ or $d_f(k)$) needs to be estimated.

Remark 4. The reasons why we present the CDO are given as: 1) in practical control systems, disturbances can destroy the performance of the systems and there are many forms of disturbances such as modeled disturbances, unmodeled disturbances, and stochastic disturbances; 2) the existing disturbance observers usually can estimate only one form of disturbance, and it is difficult for these disturbance observers to obtain accurate estimates of multi-source disturbances; 3) the presented CDO can make use of the characteristics of the modeled disturbances and the unmodeled disturbances to estimate these two kinds of disturbances accurately.

Remark 5. The novelty of the proposed CDO is summarized as: 1) the proposed CDO can not only estimate the modeled disturbances but also the norm bounded nonlinear signals, which is of great significance in practical engineering; 2) the noise-to-state stability in probability of the CDO is guaranteed by the proposed sufficient conditions which are novel and less conservative.

Remark 6. The analysis of the effects on the observer accuracy is shown as: 1) the proposed CDO can estimate the unexpected nonlinearities and the modeled disturbances separately, which is more accurate than only using the ESO to estimate the total nonlinearities directly; 2) the CDO combines the advantages of both the ESO and the DO. In the proposed CDO, the estimation of the nonlinearities is considered into the estimation of the modeled disturbance. Similarly, the estimation of the modeled disturbance is also considered into the estimation of the nonlinearities. Through this design, the estimation accuracy is greatly improved.

Remark 7. The reason why not estimate $d_s(k)$ is given as follows: by setting the bandwidth of the CDO high enough, the information of $d_s(k)$ can be included in the observer. In this case, not only the information of $d_s(k)$ will be included, but also more noise information will be included, which will lead to the decline of the observation accuracy. Therefore, the CDO is designed to get $\hat{d}_o(k)$ and $\hat{d}_f(k)$ but not the estimation of $d_s(k)$.

Before proceeding, the following definition is presented which is instrumental subsequently.

Definition 1. Jiao et al. [35] System 10 is said to be noise-to-state stable (NSS) in probability if for any $0 < \varepsilon < 1$, there exist a class $\mathcal{KL}$ function $\tilde{\tau}_2(\cdot)$ and a class $\mathcal{K}$ function $\tilde{\varsigma}_2(\cdot)$ such that for any $e_{wf}(0)$, there holds

$$\mathbf{Pr}\{\|e_{wf}\|_2^2 \le \tilde{\varsigma}_2(\cdot) + \tilde{\tau}_2(\cdot)\} \ge 1 - \epsilon$$

In the following, we provide sufficient conditions to guarantee that the error dynamics (10) is NSS in probability.

Theorem 1. *Consider error dynamics (10). If there exist positive definite matrices P_w and P_f such that the following linear matrix inequality*

$$\begin{bmatrix} -\frac{1}{1+\lambda_2}P_w & \mathbf{0}_{3\times 3} & P_wW - Q_wGV & Q_wG \\ & -\frac{1}{1+\lambda_2}P_f & Q_fGV & P_f - Q_fG \\ & * & -P_w & \mathbf{0}_{3\times 3} \\ & * & * & -P_f \end{bmatrix} < 0 \tag{11}$$

holds, where $\lambda_2 > 0$, then system (10) is NSS in probability satisfying

$$\mathbf{Pr}\left\{\|e_{wf}(k)\|_2^2 \le \zeta_2(e_{wf}(0), k) + \theta_2(\|\tilde{\eta}_1(k)\|_\infty^2)\right\} \ge 1 - \epsilon_2$$

with

$$\zeta_2(e_{wf}(0), k) = \frac{\lambda_{\max}(P_{wf})}{\epsilon_2\lambda_{\min}(P_{wf})}\left(1 - \left(1 - \frac{1}{\rho_2}\right)\frac{\hat{\alpha}_3}{\lambda_{\max}(P_{wf})}\right)^k \times \|e_{wf}(0)\|_2^2$$

$$\theta_2(\|\tilde{\eta}_1(k)\|_\infty^2) = \frac{\tau_3\lambda_{\max}(P_{wf})}{\epsilon_2\hat{\alpha}_3\lambda_{\min}(P_{wf})}\phi_1\left(\|\tilde{\eta}_1(k)\|_\infty^2\right)$$

$$\phi_1\left(\|\tilde{\eta}_1(k)\|_\infty^2\right) = \left(1 + \frac{1}{\lambda_2}\right)\|\mathbb{E}\{\tilde{\eta}_1^T(k)\tilde{\eta}_1(k)\}\|_\infty$$

$$\begin{aligned}\|\mathbb{E}\{\tilde{\eta}_1^T(k)\tilde{\eta}_1(k)\}\|_\infty = {} & \lambda_{\max}(G^TL_w^TL_wG + G^TL_f^TL_fG)\bar{d}_s^2 \\ & + \lambda_{\max}(2G^TL_w^TL_wG + G^TL_f^TL_fG)\tilde{d}_s^2 + 3\lambda_{\max}(H^TH)\bar{\eta}^2 \\ & + \lambda_{\max}(3G^TL_w^TL_wG + 2(\mathbf{I}_{3\times 3} - L_fG)^T(\mathbf{I}_{3\times 3} - L_fG))\delta_f^2\end{aligned}$$

$$0 < \hat{\alpha}_3 \le \min\left\{\varsigma_3, \lambda_{\max}(P_{wf})\right\}, \ \ \rho_2 > 1, \ \ 0 < \epsilon_2 < 1, \ \ \tau_3 = \lambda_{\max}(P_{wf})$$

$$\varsigma_3 = \lambda_{\min}(P_{wf} - (1 + \lambda_2)\Omega^TP_{wf}\Omega)$$

Furthermore, we have $L_w = P_w^{-1}Q_w$ *and* $L_f = P_f^{-1}Q_f$.

Proof. Define $P_{wf} = \begin{bmatrix} P_w & \mathbf{0}_{3\times3} \\ \mathbf{0}_{3\times3} & P_f \end{bmatrix}$ where P_w and P_f are both positive definite matrices. Select the Lyapunov function $V_{wf}(k) = e_{wf}^T(k)P_{wf}e_{wf}(k)$. It is easy to obtain

$$\begin{aligned}&\mathbb{E}\{V_{wf}(k+1)\} - \mathbb{E}\{V_{wf}(k)\}\\ =&\mathbb{E}\big\{e_{wf}^T(k)\Omega^T P_{wf}\Omega e_{wf}(k) + \tilde{\eta}_1^T(k)P_{wf}\tilde{\eta}_1(k) + e_{wf}^T(k)\Omega^T P_{wf}\tilde{\eta}_1(k)\\ &+\tilde{\eta}_1^T(k)P_{wf}\Omega e_{wf}(k)\big\} - \mathbb{E}\{V_{wf}(k)\}\end{aligned} \tag{12}$$

By Young's inequality [34], the following inequality

$$\begin{aligned}&e_{wf}^T(k)\Omega^T P_{wf}\tilde{\eta}_1(k) + \tilde{\eta}_1^T(k)P_{wf}\Omega e_{wf}(k)\\ \le&\lambda_2 e_{wf}^T(k)\Omega^T P_{wf}\Omega e_{wf}(k) + \frac{1}{\lambda_2}\tilde{\eta}_1^T(k)P_{wf}\tilde{\eta}_1(k)\end{aligned}$$

holds for any $\lambda_2 > 0$. Then, we have

$$\begin{aligned}&\mathbb{E}\{V_{wf}(k+1)\} - \mathbb{E}\{V_{wf}(k)\}\\ \le&\mathbb{E}\{e_{wf}^T(k)((1+\lambda_2)\Omega^T P_{wf}\Omega - P_{wf})e_{wf}(k)\} + \left(1+\frac{1}{\lambda_2}\right)\mathbb{E}\{\tilde{\eta}_1^T(k)P_{wf}\tilde{\eta}_1(k)\}\\ \le&-\varsigma_3\mathbb{E}\{\|e_{wf}(k)\|_2^2\} + \tau_3\phi_1\big(\|\tilde{\eta}_1(k)\|_\infty^2\big)\end{aligned}$$

According to Schur complement, if (11) holds, we have $(1+\lambda_2)\Omega^T P_{wf}\Omega - P_{wf} < 0$. Furthermore, based on (12), we have

$$\begin{aligned}&\mathbb{E}\{V_{wf}(k+1)\} - \mathbb{E}\{V_{wf}(k)\}\\ \le&-\frac{\hat{\alpha}_3}{\lambda_{\max}(P_{wf})}\mathbb{E}\big\{V_{wf}(k)\big\} + \tau_3\phi_1\big(\|\tilde{\eta}_1(k)\|_\infty^2\big)\end{aligned}$$

where $0 < \hat{\alpha}_3 \le \min\big\{\varsigma_3, \lambda_{\max}(P_{wf})\big\}$. Thus, we have

$$\mathbb{E}\{V_{wf}(k+1)\} \le\left(1-\frac{\hat{\alpha}_3}{\lambda_{\max}(P_{wf})}\right)\mathbb{E}\big\{V_{wf}(k)\big\} + \tau_3\phi_1\big(\|\tilde{\eta}_1(k)\|_\infty^2\big) \tag{13}$$

For any $\rho_2 > 1$, we define

$$\mathcal{B} = \big\{e_{wf}(k) : \mathbb{E}\{V_{wf}(k)\} \le \theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)\big\}$$

in which $\theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2) = \frac{\lambda_{\max}(P_{wf})\rho_2\tau_3}{\hat{\alpha}_3}\phi_1\big(\|\tilde{\eta}_1(k)\|_\infty^2\big)$. In the following, we aim to prove that, for any positive scalar $\epsilon_2 \in (0,1)$, there exist functions $\zeta_2 \in \mathcal{KL}$ and $\theta_2 \in \mathcal{K}$ [35] such that

$$\mathbf{Pr}\big\{\|e_{wf}(k)\|_2^2 \le \zeta_2(e_{wf}(0), k) + \theta_2(\|\tilde{\eta}_1(k)\|_\infty^2))\big\} \ge 1-\epsilon_2$$

To begin with, we consider the two mutually exclusive cases: Case 1: $e_{wf}(0) \in \mathcal{B}$ and Case 2: $e_{wf}(0) \notin \mathcal{B}$. For Case 1, it follows from $e_{wf}(0) \in \mathcal{B}$ that

$$\mathbb{E}\{V_{wf}(k)\} \le \theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2) \tag{14}$$

If follows from (13) and (14) that

$$\mathbb{E}\{V_{wf}(1)\} \le\left(1-\frac{\hat{\alpha}_3}{\lambda_{\max}(P_{wf})}\right)\mathbb{E}\big\{V_{wf}(0)\big\}$$

$$+\tau_3\phi_1\left(\|\tilde{\eta}_1(k)\|_\infty^2\right) < \theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)$$

In virtue of the mathematical induction, one has, for any $k \in \mathbb{Z}^+$, $\mathbb{E}\{V_{wf}(1)\} \leq \theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)$. Applying Chebyshev's inequality yields

$$\mathbf{Pr}\left\{V_{wf}(k) \geq \tfrac{1}{\epsilon_2}\theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)\right\} \leq \tfrac{\mathbb{E}\{V_{wf}(k)\}\epsilon_2}{\theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)} \leq \epsilon_2$$

It is easy to obtain

$$\begin{aligned}&\mathbf{Pr}\left\{\lambda_{\min}(P_{wf})\|e_{wf}(k)\|_2^2 \geq \frac{1}{\epsilon_2}\theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)\right\}\\ \leq&\mathbf{Pr}\left\{V_{wf}(k) \geq \frac{1}{\epsilon_2}\theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)\right\} \leq \epsilon_2\end{aligned}$$

which indicates

$$\mathbf{Pr}\left\{\|e_{wf}(k)\|_2^2 \leq \frac{1}{\lambda_{\min}(P_{wf})\epsilon_2}\theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)\right\} \geq 1-\epsilon_2$$

Letting $\rho_2 \to 1^+$, we have, for any $k \in \mathbb{Z}^+$,

$$\mathbf{Pr}\left\{\|e_{wf}(k)\|_2^2 \leq \theta_2(\|\tilde{\eta}_1(k)\|_\infty^2)\right\} \geq 1-\epsilon_2 \tag{15}$$

where $\theta_2(\|\tilde{\eta}_1(k)\|_\infty^2) = \frac{\tau_3\lambda_{\max}(P_{wf})}{\epsilon_2\hat{\alpha}_3\lambda_{\min}(P_{wf})}\phi_1\left(\|\tilde{\eta}_1(k)\|_\infty^2\right)$. Therefore, for any $\zeta_2 \in \mathcal{KL}$ [35], we have

$$\begin{aligned}&\mathbf{Pr}\left\{\|e_{wf}(k)\|_2^2 \leq \theta_2(\|\tilde{\eta}_1(k)\|_\infty^2)\right\}\\ \leq&\mathbf{Pr}\left\{\|e_{wf}(k)\|_2^2 \leq \zeta_2(e_{wf}(0), k) + \theta_2(\|\tilde{\eta}_1(k)\|_\infty^2)\right\}\end{aligned} \tag{16}$$

According to (15) and (16), Case 1 can be proven. For Case 2, we have $\mathbb{E}\{V_{wf}(0)\} > \theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)$. In what follows, two subcases are considered: Subcase 2.1: $\mathbb{E}\{V_{wf}(k)\} > \theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)$ for all $k \in \mathbb{Z}^+$; Subcase 2.2: $\mathbb{E}\{V_{wf}(k)\} > \theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)$ for all $k \in [1, k^*-1]$ and $\mathbb{E}\{V_{wf}(k)\} \leq \theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)$ for $k \geq k^*$ where

$$k^* = \min\left\{k : \mathbb{E}\{V_{wf}(k)\} \leq \theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)\right\}$$

If Subcase 2.1 holds, then we have

$$\tau_3\phi_1\left(\|\tilde{\eta}_1(k)\|_\infty^2\right) < \frac{1}{\rho_2}\frac{\hat{\alpha}_3}{\lambda_{\max}(P_{wf})}\mathbb{E}\{V_{wf}(k)\} \tag{17}$$

for all $k \in \mathbb{Z}^+$. Substituting (17) into (13), we have

$$\mathbb{E}\left\{V_{wf}(k+1)\right\} - \mathbb{E}\left\{V_{wf}(k)\right\} \leq \left(\frac{1}{\rho_2}-1\right)\frac{\hat{\alpha}_3}{\lambda_{\max}(P_{wf})}\mathbb{E}\{V_{wf}(k)\}$$

Noting that $\rho_2 > 1$ and applying the standard comparison lemma in [36], we conclude that there exists a function $\hat{\zeta}_2 \in \mathcal{KL}$ such that $\mathbb{E}\{V_{wf}(k))\} \leq \hat{\zeta}_2(V_{wf}(0)), k)$ where

$$\hat{\zeta}_2(V_{wf}(0), k) = \left(1-\left(1-\frac{1}{\rho_2}\right)\frac{\hat{\alpha}_3}{\lambda_{\max}(P_{wf})}\right)^k\mathbb{E}\{V_{wf}(0)\}$$

for all $k \in \mathbb{Z}^+$. Resorting to Chebyshev's inequality again, we have

$$\mathbf{Pr}\left\{V_{wf}(k) \geq \frac{1}{\epsilon_2}\hat{\zeta}_2(V_{wf}(0), k)\right\} \leq \frac{\mathbb{E}\{V_{wf}(k)\}\epsilon_2}{\hat{\zeta}_2(V_{wf}(0), k)} \leq \epsilon_2$$

It is easy to obtain

$$\begin{aligned}&\mathbf{Pr}\left\{\lambda_{\min}(P_{wf})\|e_{wf}(k)\|_2^2 \geq \frac{1}{\epsilon_2}\hat{\zeta}_2(V_{wf}(0),k)\right\}\\ \leq&\mathbf{Pr}\left\{V_{wf}(k) \geq \frac{1}{\epsilon_2}\hat{\zeta}_2(V_{wf}(0),k)\right\}\end{aligned}$$

Since $\hat{\zeta}_2$ is a $\mathcal{KL}$-function [35], we have

$$\begin{aligned}&\mathbf{Pr}\left\{\lambda_{\min}(P_{wf})\|e_{wf}(k)\|_2^2 \geq \frac{\hat{\zeta}_2(\lambda_{\max}(P_{wf})\|e_{wf}(0)\|_2^2,k)}{\epsilon_2}\right\}\\ \leq&\mathbf{Pr}\left\{\lambda_{\min}(P_{wf})\|e_{wf}(k)\|_2^2 \geq \frac{1}{\epsilon_2}\hat{\zeta}_2(V_{wf}(0),k)\right\}\end{aligned}$$

Therefore, one has

$$\mathbf{Pr}\left\{\|e_{wf}(k)\|_2^2 \geq \frac{\hat{\zeta}_2(\lambda_{\max}(P_{wf})\|e_{wf}(0)\|_2^2,k)}{\epsilon_2\lambda_{\min}(P_{wf})}\right\} \leq \epsilon_2$$

for all $k \in \mathbb{Z}^+$. Denote

$$\zeta_2(e_{wf}(0),k) = \frac{\lambda_{\max}(P_{wf})}{\epsilon_2\lambda_{\min}(P_{wf})}\left(1-\left(1-\frac{1}{\rho_2}\right)\frac{\hat{\alpha}_3}{\lambda_{\max}(P_{wf})}\right)^k\|e_{wf}(0)\|_2^2$$

Then, we have $\mathbf{Pr}\{\|e_{wf}(k)\|_2^2 \leq \zeta_2(e_{wf}(0),k)\} \geq 1-\epsilon_2$ which further results in

$$\mathbf{Pr}\{\|e_{wf}(k)\|_2^2 \leq \zeta_2(e_{wf}(0),k) + \theta_2(\|\tilde{\eta}_1(k)\|_\infty^2)\} \geq 1-\epsilon_2 \tag{18}$$

for all $k \in \mathbb{Z}^+$. On the other hand, if Subcase 2.2 holds, then we have $\mathbb{E}\{V_{wf}(k)\} > \theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)$ for all $k \in [0, k^*-1]$, which leads to

$$\mathbf{Pr}\{\|e_{wf}(k)\|_2^2 \leq \zeta_2(e_{wf}(0),k)\} \geq 1-\epsilon_2 \tag{19}$$

When $k = k^*$, it is readily obtained from the definition of k^* that $\mathbb{E}\{V_{wf}(k^*)\} \leq \theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)$ which is followed immediately by $\mathbb{E}\{V_{wf}(k)\} \leq \theta_2^*(\rho_2, \|\tilde{\eta}_1(k)\|_\infty^2)$ for any $k \geq k^*$ by following the same procedure as (14). Repeating the proof process of Case 1, we have

$$\mathbf{Pr}\left\{\sup_{k\geq k^*}\|e_{wf}(k)\|_2^2 \leq \theta_2(\|\tilde{\eta}_1(k)\|_\infty^2)\right\} > 1-\epsilon_2 \tag{20}$$

Combining (19) with (20) indicates that, for all $k \in \mathbb{Z}^+$, we have

$$\mathbf{Pr}\{\|e_{wf}(k)\|_2^2 \leq \zeta_2(e_{wf}(0),k) + \theta_2(\|\tilde{\eta}_1(k)\|_\infty^2)\} \geq 1-\epsilon_2 \tag{21}$$

According to (18) and (21), Case 2 can be proven. The proof is complete. □

Remark 8. In Theorem 3.1, the CDO gain can be obtained by solving the LMI (12). Furthermore, the noise-to-state stability in probability of the estimation error is established. The The noise-to-state stability in probability is a kind of stability from time zero to infinity, which inflects transient performance and steady performance. From Theorem 3.1, it is easy to find the transient performance $\zeta_2(e_{wf}(0),k)$ and steady performance $\theta_2(\|\tilde{\eta}_1(k)\|_\infty^2)$. As $k \to +\infty$, the transient part $\zeta_2(e_{wf}(0),k) \to 0$.

Remark 9. Disturbance observers have been widely studied because of their excellent performance [10,11,40–43]. Compared with these studies, the novelty of the CDO is clarified as: 1) the CDO can not only observe the estimates of disturbances generated by exogenous systems, but also enables to observe the estimate of unknown nonlinearities; 2) the noise-to-state in probability of the proposed CDO is less conservative.

4. Composite stochastic model predictive control scheme

In this section, we propose the C-SMPC scheme to generate the input signal for system (7). In the first subsection, the composite control strategy is presented, and the probabilistic constraints are reformulated as the deterministic constraints. In the second subsection, the SMPC part is designed to generate the optimal control signal $\bar{u}^*(k)$ by solving an optimization problem. Finally, the feasibility and stability of system (7) under the proposed composite strategy are analyzed.

4.1. Reformulation of the probabilistic constraints

Consider the following nominal system of (7) as

$$\bar{x}(k+1) = A\bar{x}(k) + B\bar{u}(k) \tag{22}$$

where $\bar{x}(k)$ is the nominal state and $\bar{u}(k)$ is the nominal input. Furthermore, the C-SMPC strategy is designed as

$$u(k) = \bar{u}(k) + \tilde{e}_u(k) \tag{23}$$

$$\tilde{e}_u(k) = K(x(k) - \bar{x}(k)) - \hat{d}_f(k) - \hat{d}_o(k) \tag{24}$$

where $\bar{u}(k) = \bar{u}^*(k)$ if $x(k) \notin \mathcal{X}_f$, and $\bar{u}(k) = K\bar{x}(k)$ if $x(k) \in \mathcal{X}_f$. Note that $\bar{u}^*(k)$ is obtained by solving an optimization problem shown subsequently. K is the controller parameter and $\mathcal{X}_f$ is the terminal region. From (23) and (24), it is obvious that the input consists of the tube MPC part: $\bar{u}(k) + K(x(k) - \bar{x}(k))$ and the disturbances rejection part: $-\hat{d}_f(k) - \hat{d}_o(k)$. Note that $\bar{u}(k) = \bar{u}^*(k)$ is calculated by solving an optimization problem when $x(k) \notin \mathcal{X}_f$ to move the state $x(k)$ along a tube whose center is $\bar{x}(k)$, while $K(x(k) - \bar{x}(k))$ is used to keep the state $x(k)$ within this tube.

We rewrite system (7) as

$$x(k) = \bar{x}(k) + \tilde{e}_x(k) \tag{25}$$

$$\begin{aligned}\tilde{e}_x(k+1) = {} & A\tilde{e}_x(k) + BK(x(k) - \bar{x}(k)) + Be_f(k) \\ & + BVe_w(k) + Gd_s(k) + GO(T^2)\end{aligned} \tag{26}$$

where $\tilde{e}_x(0) = \mathbf{0}_{6\times 1}$. According to (23) and (25), we have $\mathbb{E}\{x(k)\} = \bar{x}(k) + \mathbb{E}\{\tilde{e}_x(k)\}$ and $\mathbb{E}\{u(k)\} = \bar{u}(k) + \mathbb{E}\{\tilde{e}_u(k)\}$ which implies $x(k) - \mathbb{E}\{x(k)\} = x(k) - \bar{x}(k) - \mathbb{E}\{\tilde{e}_x(k)\}$ and $u(k) - \mathbb{E}\{u(k)\} = u(k) - \bar{u}(k) - \mathbb{E}\{\tilde{e}_u(k)\}$. Then, we have

$$\begin{aligned}& \mathbb{E}\left\{(x(k) - \mathbb{E}\{x(k)\})(x(k) - \mathbb{E}\{x(k)\})^T\right\} \\ =& \mathbb{E}\left\{(x(k) - \bar{x}(k))(x(k) - \bar{x}(k))^T\right\} - \mathbb{E}\{\tilde{e}_x(k)\}\mathbb{E}\left\{\tilde{e}_x^T(k)\right\}\end{aligned} \tag{27}$$

$$\begin{aligned}&\mathbb{E}\left\{(u(k)-\mathbb{E}\{u(k)\})(u(k)-\mathbb{E}\{u(k)\})^T\right\}\\=&\mathbb{E}\left\{(u(k)-\bar{u}(k))(u(k)-\bar{u}(k))^T\right\}-\mathbb{E}\{\tilde{e}_u(k)\}\mathbb{E}\left\{\tilde{e}_u^T(k)\right\}\end{aligned}\tag{28}$$

Defining $\mu_x(k)=x(k)-\bar{x}(k)$, we have $\mu_x(k+1)=\Phi_x\mu_x(k)+Ge_f(k)+GVe_w(k)+Gd_s(k)+GO(T^2)$ with $\Phi_x=A+BK$. Similarly, we have

$$\begin{aligned}&\mathbb{E}\{\mu_x(k+1)\mu_x^T(k+1)\}\\&\le(1+\tilde{\lambda}_1+\tilde{\lambda}_2+\tilde{\lambda}_3+\tilde{\lambda}_4)\Phi_x\mathbb{E}\{\mu_x(k)\mu_x^T(k)\}\Phi_x^T\\&+G\mathbb{E}\{d_s(k)d_s^T(k)\}G^T+(3+1/\tilde{\lambda}_4)\tilde{d}_s^2GG^T+(4+1/\tilde{\lambda}_3)\delta_f^2GG^T\\&+(4+1/\tilde{\lambda}_1)\|e_{wf}(k)\|_2^2GG^T+(4+1/\tilde{\lambda}_2)\|e_{wf}(k)\|_2^2GVV^TG^T\end{aligned}$$

where

$$\mathbb{E}\{d_s(k)d_s^T(k)\}=\begin{bmatrix}\breve{d}_{s,1}+\tilde{d}_{s,1}^2 & \tilde{d}_{s,1}\tilde{d}_{s,2} & \tilde{d}_{s,1}\tilde{d}_{s,3}\\ \tilde{d}_{s,2}\tilde{d}_{s,1} & \breve{d}_{s,2}+\tilde{d}_{s,2}^2 & \tilde{d}_{s,2}\tilde{d}_{s,3}\\ \tilde{d}_{s,3}\tilde{d}_{s,1} & \tilde{d}_{s,3}\tilde{d}_{s,2} & \breve{d}_{s,3}+\tilde{d}_{s,3}^2\end{bmatrix}$$

and $\tilde{\lambda}_1>0$, $\tilde{\lambda}_2>0$, $\tilde{\lambda}_2>0$, $\tilde{\lambda}_4>0$ are constants satisfying

$$\lambda\left((1+\tilde{\lambda}_1+\tilde{\lambda}_2+\tilde{\lambda}_3+\tilde{\lambda}_4)^{0.5}\Phi_x\right)\in(-1,1)$$

Denoting

$$\begin{aligned}&\hat{\Pi}_x(k+1)=(1+\tilde{\lambda}_1+\tilde{\lambda}_2+\tilde{\lambda}_3+\tilde{\lambda}_4)\Phi_x\mathbb{E}\{\mu_x(k)\mu_x^T(k)\}\Phi_x^T\\&+G\mathbb{E}\{d_s(k)d_s^T(k)\}G^T+(4+1/\tilde{\lambda}_1)\|e_{wf}(k)\|_2^2GG^T\\&+(4+1/\tilde{\lambda}_2)\|e_{wf}(k)\|_2^2GVV^TG^T+(4+1/\tilde{\lambda}_3)\delta_f^2GG^T\\&+(3+1/\tilde{\lambda}_4)\tilde{d}_s^2GG^T+(4+1/\tilde{\lambda}_3)\delta_f^2GG^T+(4+1/\tilde{\lambda}_3)\delta_f^2GG^T\end{aligned}$$

with $\hat{\Pi}_x(0)=\mathbf{0}_{6\times6}$, we have $\mathbb{E}\left\{\mu_x(k)\mu_x^T(k)\right\}\le\hat{\Pi}_x(k)$ which is

$$\mathbb{E}\left\{(x(k)-\bar{x}(k))(x(k)-\bar{x}(k))^T\right\}=\mathbb{E}\left\{\mu_x(k)\mu_x^T(k)\right\}\le\hat{\Pi}_x(k)$$

From (27), we have

$$\begin{aligned}&\mathbb{E}\left\{(x(k)-\mathbb{E}\{x(k)\})(x(k)-\mathbb{E}\{x(k)\})^T\right\}\\&\le\hat{\Pi}_x(k)-\mathbb{E}\{\tilde{e}_x(k)\}\mathbb{E}\left\{\tilde{e}_x^T(k)\right\}\end{aligned}\tag{29}$$

It follows from (26) that $\mathbb{E}\{\tilde{e}_x(k)\}=\sum_{i=0}^{k}(A+BK)^{i-1}\tilde{\omega}_x(i)$ with

$$\begin{aligned}\tilde{\omega}_x(k)&=Be_f(k)+BVe_w(k)+G\tilde{d}_s+GO(T^2)\\&=\tilde{B}_1e_{wf}(k)+G\tilde{d}_s+GO(T^2)\end{aligned}$$

$\tilde{B}_1=[BV\ \ B]$

Furthermore, we have

$$\begin{aligned}&\mathbb{E}\{(u(k)-\bar{u}(k))(u(k)-\bar{u}(k))^T\}\\&\le3K\hat{\Pi}_x(k)K^T+6\bar{d}_f^2\mathbf{I}_{3\times3}+6\|e_{wf}(k)\|_2^2\mathbf{I}_{3\times3}\end{aligned}$$

$$+ 6\|e_{wf}(k)\|_2^2 VV^T + 6\bar{w}^2 VV^T$$

Denoting

$$\begin{aligned}\hat{\Pi}_u(k) =& 3K\hat{\Pi}_x(k)K^T + 6\bar{d}_f^2\mathbf{I}_{3\times3} + 6\|e_{wf}(k)\|_2^2\mathbf{I}_{3\times3}\\ &+ 6\|e_{wf}(k)\|_2^2 VV^T + 6\bar{w}^2 VV^T\end{aligned}$$

we have $\mathbb{E}\{(u(k)-\bar{u}(k))(u(k)-\bar{u}(k))^T\} \le \hat{\Pi}_u(k)$. According to (28), we have

$$\begin{aligned}&\mathbb{E}\left\{(u(k)-\mathbb{E}\{u(k)\})(u(k)-\mathbb{E}\{u(k)\})^T\right\}\\ \le& \hat{\Pi}_u(k) - \mathbb{E}\{\tilde{e}_u(k)\}\mathbb{E}\left\{\tilde{e}_u^T(k)\right\}\end{aligned} \tag{30}$$

From the above steps, inequalities (29) and (30) are obtained. By these inequalities, probabilistic constraints (8) and (9) can be reformulated as deterministic constraints in the following lemma.

Lemma 1. *Consider system (7) with the probabilistic constraints (8) and (9). The probabilistic constraints (8) and (9) can be reformulated as*

$$b_x^T\bar{x}(k) \le h_x - ((1-l_x)\sqrt{1/l_x} + \sqrt{l_x})\sqrt{b_x^T\hat{\Pi}_x(k)b_x} \tag{31a}$$

$$b_u^T\bar{u}(k) \le h_u - ((1-l_u)\sqrt{1/l_u} + \sqrt{l_u})\sqrt{b_u^T\hat{\Pi}_u(k)b_u} \tag{31b}$$

Proof. Denote $X(k) = \mathbb{E}\left\{(x(k)-\mathbb{E}\{x(k)\})(x(k)-\mathbb{E}\{x(k)\})^T\right\}$. According to Cantelli's inequality [37], we have

$$\mathbf{Pr}\{b_x^T x(k) \ge h_x\} \le \mathbf{Pr}\{b_x^T x(k) \ge b_x^T\mathbb{E}\{x(k)\} + \Delta\} \le \frac{b_x^T X(k) b_x}{b_x^T X(k) b_x + \Delta^2} \tag{32}$$

Based on (29) and (32), we have

$$\begin{aligned}&\mathbf{Pr}\{b_x^T x(k) \ge h_x\} \le \mathbf{Pr}\left\{b_x^T x(k) \ge \frac{b_x^T X(k) b_x}{b_x^T X(k) b_x + \Delta^2}\right\}\\ \le& \frac{b_x^T\left(\hat{\Pi}_x(k) - \mathbb{E}\{\tilde{e}(k)\}\mathbb{E}\left\{\tilde{e}^T(k)\right\}\right)b_x}{b_x^T\left(\hat{\Pi}_x(k) - \mathbb{E}\{\tilde{e}(k)\}\mathbb{E}\left\{\tilde{e}^T(k)\right\}\right)b_x + \Delta^2}\end{aligned}$$

If $\Delta \ge \sqrt{b_x^T(\hat{\Pi}_x(k) - \mathbb{E}\{\tilde{e}(k)\}\mathbb{E}\left\{\tilde{e}^T(k)\right\})b_x}\sqrt{(1-l_x)/l_x}$ holds, (8) holds. It follows from $b_x^T\mathbb{E}\{x(k)\} \le h_x - \Delta$ that

$$b_x^T\mathbb{E}\{x(k)\} \le h_x - \sqrt{b_x^T(\hat{\Pi}_x(k) - \mathbb{E}\{\tilde{e}(k)\}\mathbb{E}\left\{\tilde{e}^T(k)\right\})b_x}\sqrt{(1-l_x)/l_x}$$

which implies

$$\begin{aligned}b_x^T\bar{x}(k) \le& -\sqrt{b_x^T(\hat{\Pi}_x(k) - \mathbb{E}\{\tilde{e}(k)\}\mathbb{E}\left\{\tilde{e}^T(k)\right\})b_x}\sqrt{(1-l_x)/l_x}\\ &+ h_x - b_x^T\mathbb{E}\{\tilde{e}(k)\}\end{aligned}$$

Define the function

$$F(b_x^T\mathbb{E}\{\tilde{e}(k)\}) = -\sqrt{b_x^T(\hat{\Pi}_x(k) - \mathbb{E}\{\tilde{e}(k)\}\mathbb{E}\left\{\tilde{e}^T(k)\right\})b_x}\sqrt{(1-l_x)/l_x}$$

$$- b_x^T \mathbb{E}\{\tilde{e}(k)\} \tag{33}$$

For simplicity, we define $a = b_x^T \mathbb{E}\{\tilde{e}(k)\}$, $b = b_x^T \hat{\Pi}_x(k) b_x$ and $c = \sqrt{(1-l_x)/l_x}$. Then, (33) can be reformulated as $F(a) = -c\sqrt{b-a^2} - a$. It is easy to see that the function $F(a) = -c\sqrt{b-a^2} - a$ achieves the minimum at $a = \sqrt{b/(c^2+1)}$ with the minimal value $h_x - ((1-l_x)\sqrt{1/l_x} + \sqrt{l_x})\sqrt{b_x^T \hat{\Pi}_x(k) b_x}$. Therefore, we have

$$\begin{aligned} b_x^T \bar{x}(k) \leq & h_x - ((1-l_x)\sqrt{1/l_x} + \sqrt{l_x})\sqrt{b_x^T \hat{\Pi}_x(k) b_x} \\ \leq & -\sqrt{b_x^T (\hat{\Pi}_x(k) - \mathbb{E}\{\tilde{e}(k)\}\mathbb{E}\{\tilde{e}^T(k)\}) b_x}\sqrt{(1-l_x)/l_x} \\ & + h_x - b_x^T \mathbb{E}\{\tilde{e}(k)\} \end{aligned}$$

Similarly, we have

$$\begin{aligned} b_u^T \bar{u}(k) \leq & h_u - ((1-l_u)\sqrt{1/l_u} + \sqrt{l_u})\sqrt{b_u^T \hat{\Pi}_u(k) b_u} \\ \leq & -\sqrt{b_u^T (\hat{\Pi}_u(k) - \mathbb{E}\{e_u(k)\}\mathbb{E}\{e_u^T(k)\}) b_u}\sqrt{(1-l_u)/l_u} \\ & + h_u - b_u^T \mathbb{E}\{e_u(k)\} \end{aligned}$$

The proof is complete. □

Remark 10. In Lemma 4.1, the probabilistic constraints (8) and (9) are reformulated as the deterministic constraints (31a) and (31b), respectively. By substituting (31a) and (31b) with (8) and (9) in the optimization problem, we can obtain the optimal feedback input signal.

Remark 11. The reasons why we present the MPC are given as: 1) state and input constraints exist widely in control systems, and MPC can handle these constraints effectively; 2) MPC is an optimization algorithm, which can help the system to achieve good performance.

4.2. Stochastic model predictive control scheme

In this subsection, the SMPC scheme is designed to generate input signal for the system (7). Based on the knowledge (e.g., measurements or estimates) available at time k, the optimal input signal $\bar{u}^*(k)$ is calculated. The cost function to be minimized is

$$\begin{aligned} J(\bar{x}(k), \bar{u}(k)) = & \sum_{i=0}^{N-1} \left(\|\bar{x}(k+i|k)\|_Q^2 + \|\bar{u}(k+i|k)\|_R^2 \right) \\ & + \|\bar{x}(k+N|k)\|_P^2 \end{aligned} \tag{34}$$

where $\bar{x}(k+i|k)$ is the predicted state which is also the nominal state with $\bar{x}(k|k) = \bar{x}(k)$; $\bar{u}(k+i|k)$ is the predicted input which is also the input of the nominal system; $N \in \mathbb{Z}^+$ is the prediction horizon; $Q > 0$, $R > 0$ and $P > 0$ are weighting matrices with appropriate dimensions.

To guarantee recursive feasibility and stability, some terminal constraints must be considered. In this paper, the terminal constraints are given by

$$\bar{x}(k+N) \in \mathcal{X}_f \tag{35a}$$

$$\hat{\Pi}_x(k+N) \leq \bar{\Pi}_{xx} \tag{35b}$$

where $\mathcal{X}_f$ is a positively invariant set to be designed such that we have

$$(A+BK)\bar{x} \in \mathcal{X}_f, \ \ \forall \bar{x} \in \mathcal{X}_f \tag{36}$$

Note that $\bar{\Pi}_{xx}$ is obtained by solving the following Lyapunov equation:

$$\begin{aligned}\bar{\Pi}_{xx} =& \mathbb{E}\{\mu_x\mu_x^T\} = (1+\tilde{\lambda}_1+\tilde{\lambda}_2+\tilde{\lambda}_3+\tilde{\lambda}_4)\Phi_x\mathbb{E}\{\mu_x\mu_x^T\}\Phi_x^T \\ &+ G\mathbb{E}\{d_s(k)d_s^T(k)\}G^T + (4+1/\tilde{\lambda}_1)\|e_{wf}(k)\|_2^2 GG^T \\ &+ (4+1/\tilde{\lambda}_2)\|e_{wf}(k)\|_2^2 GVV^TG^T \\ &+ (4+1/\tilde{\lambda}_3)\delta_f^2 GG^T + (3+1/\tilde{\lambda}_4)\tilde{d}_s^2 GG^T \\ \bar{\Pi}_{uu} =& 3K\bar{\Pi}_{xx}K^T + 6\bar{d}_f^2\mathbf{I}_{3\times3} + 6\|e_{wf}(k)\|_2^2\mathbf{I}_{3\times3} \\ &+ 6\|e_{wf}(k)\|_2^2 VV^T + 6\bar{w}^2VV^T\end{aligned} \tag{37}$$

In addition, according to (31), the following conditions should be satisfied

$$b_x^T\bar{x}(k+N) \leq h_x - ((1-l_x)\sqrt{1/l_x}+\sqrt{l_x})\sqrt{b_x^T\bar{\Pi}_{xx}b_x} \tag{38a}$$

$$b_u^T\bar{u}(k+N) \leq h_u - ((1-l_u)\sqrt{1/l_u}+\sqrt{l_u})\sqrt{b_u^T\bar{\Pi}_{uu}b_u} \tag{38b}$$

for all $\bar{x} \in \mathcal{X}_f$. Considering these terminal constraints as well as constraints (31a) and (31b), the SMPC problem is formulated as follows:

Problem 1: $\bar{u}^*(k+i|k) = \arg\min_{\bar{u}(k+i|k)} J(\bar{x}(k+i|k), \bar{u}(k+i|k))$ subject to

$$\bar{x}(k+i+1|k) = A\bar{x}(k+i|k) + B\bar{u}(k+i|k) \tag{39a}$$

$$b_x^T\bar{x}(k+i|k) \leq h_x - ((1-l_x)\sqrt{1/l_x}+\sqrt{l_x})\sqrt{b_x^T\hat{\Pi}_x(k+i|k)b_x} \tag{39b}$$

$$b_u^T\bar{u}(k+i|k) \leq h_u - ((1-l_u)\sqrt{1/l_u}+\sqrt{l_u})\sqrt{b_u^T\hat{\Pi}_u(k+i|k)b_u} \tag{39c}$$

$$\bar{x}(k+N|k) \in \mathcal{X}_f \tag{39d}$$

$$\hat{\Pi}_x(k+N|k) \leq \bar{\Pi}_{xx} \tag{39e}$$

In the following lemma, we provide sufficient conditions to derive controller parameter K and guarantee that the terminal region $\mathcal{X}_f$ is an invariant region for system (7).

Lemma 2. *If* $\xi_x \geq \frac{\lambda_{\max}^{1/2}(P)(\|\tilde{B}_1\|_2\|e_{wf}(0)\|_2+\|G\|_2\bar{d}_s+\|G\|_2\delta_f)}{\lambda_{\min}(\hat{Q}_p)}$ *and*

$$\begin{bmatrix} -\Xi & \Xi A^T+\Upsilon B^T & \Upsilon & \Xi \\ & -\Xi & \mathbf{0}_{6\times3} & \mathbf{0}_{6\times6} \\ & * & -R^{-1} & \mathbf{0}_{3\times6} \\ & * & * & -Q^{-1} \end{bmatrix} < 0 \tag{40}$$

hold at the same time, where $\Xi = P^{-1}$, $\Upsilon = \Xi K^T$, $\hat{Q}_p = P^{-1/2}(Q + K^T RK)P^{-1/2}$, $\tilde{B}_1 = [BV\ B]$, *then we have* $K = \Upsilon^T \Xi^{-1}$ *and the terminal region* $\mathcal{X}_f = \left\{x(k) : x^T(k)Px(k) \le \xi_x^2\right\}$ *is an invariant region for system (7), that is,* $x(k)$ *will not depart from* $\mathcal{X}_f$ *once* $x(k)$ *enters* $\mathcal{X}_f$.

Proof. Consider the case where the states of system (7) have been driven into the terminal region $\mathcal{X}_f$ under the input (23), i.e.,

$$\mathcal{X}_f = \left\{x(k) : x^T(k)Px(k) \le \xi_x^2\right\},\ \xi_x > 0$$

According to (23), $\bar{u}(k) = K\bar{x}(k)$ is applied so that $u(k) = K\bar{x}(k) + K(x(k) - \bar{x}(k)) - \hat{d}_f(k) - \hat{d}_o(k) = Kx(k) - \hat{d}_f(k) - \hat{d}_o(k)$ will be used once $x(k)$ enters the terminal region $\mathcal{X}_f$. It is easy to obtain the closed-loop system $x(k+1) = \Phi_x x(k) + \omega_x(k)$ with $\omega_x(k) = Ge_f(k) + GVe_w(k) + Gd_s(k) + GO(T^2)$.

Select the Lyapunov function as $\tilde{V}(k) = x^T(k)Px(k)$ where P is a positive definite matrix which is also used as the penalty matrix in the cost function (34). We have $\tilde{V}(k+1) - \tilde{V}(k) = x^T(k)(\Phi_x^T P\Phi_x - P)x(k) + 2x^T P\omega_x(k)$. According to Schur complement, if (40) holds, we have $\Phi_x^T P\Phi_x - P \le -Q - K^T RK$. Furthermore, we have

$$\begin{aligned}\tilde{V}(k+1) - \tilde{V}(k) &\le -x^T(k)(Q + K^T RK)x(k) + 2x^T P\omega_x(k)\\ &\le -x^T(k)(Q + K^T RK)x(k)\\ &\quad\times\left(1 - \frac{\sqrt{\omega_x^T(k)P\omega_x(k)}}{\lambda_{\min}(\hat{Q}_p)\sqrt{x^T(k)Px(k)}}\right)\end{aligned}$$

where $\hat{Q}_p = P^{-1/2}(Q + K^T RK)P^{-1/2}$. Therefore, $\tilde{V}(k+1) \le \tilde{V}(k)$ holds if

$$\sqrt{\omega_x^T(k)P\omega_x(k)} \le \lambda_{\min}(\hat{Q}_p)\sqrt{x^T(k)Px(k)} \le \lambda_{\min}(\hat{Q}_p)\xi_x \tag{41}$$

On the other hand, we have

$$\begin{aligned}&\sqrt{\omega_x^T(k)P\omega_x(k)} \le \lambda_{\max}^{1/2}(P)\|\omega_x(k)\|_2\\ \le&\lambda_{\max}^{1/2}(P)(\|\tilde{B}_1\|_2\|e_{wf}(0)\|_2 + \|G\|_2\bar{d}_s + \|G\|_2\delta_f)\end{aligned}$$

If $\frac{\lambda_{\max}^{1/2}(P)(\|\tilde{B}_1\|_2\|e_{wf}(0)\|_2 + \|G\|_2\bar{d}_s + \|G\|_2\delta_f)}{\lambda_{\min}(\hat{Q}_p)}$ holds for all $k \in \mathbb{Z}^+$, then $x(k)$ will not depart from $\mathcal{X}_f$ once $x(k)$ enters the terminal region $\mathcal{X}_f$. The proof is complete. □

Remark 12. In Lemma 4.2, we give the invariant region of the SMPC scheme and the controller gain K. The invariant region $\mathcal{X}_f$ suggests that $x(k)$ will not depart from $\mathcal{X}_f$ once $x(k)$ enters $\mathcal{X}_f$.

Remark 13. The advantages of the SMPC are shown as: 1) it is guaranteed that the constraints with random parameters can be satisfied with a certain probability on the optimal value; 2) compared with the robust model predictive control, the SMPC can deal with stochastic system parameters and stochastic uncertainties which are unbounded.

4.3. Feasibility and stability analysis

In the part, we provide the sufficient conditions to guarantee the recursive feasibility and closed-loop stability of system (7) under the input (23).

Theorem 2. *If there exists a solution to Problem 1 at $k = 0$, then it is recursively feasible and the probabilistic constraints (8) and (9) are satisfied for all $k \in \mathbb{Z}^+ \cup \{0\}$. Furthermore, the system (7) under the input (23) is exponentially ultimately bounded in the mean square sense [38], i.e., $\mathbb{E}\{\|x(k)\|_2^2\} \le \tilde{\zeta}_3(x(0), k) + \tilde{\theta}_3$ where*

$$\tilde{\zeta}_3(x(0), k) = \frac{2}{\beta_1}\left(1 - \frac{\lambda_{\min}(Q)}{\beta_2}\right)^k J(\bar{x}^*(0), \bar{u}^*(0))$$

$$\tilde{\theta}_3 = 2(1-\beta)^{-2}\sum_{i=0}^{N}\sum_{j=0}^{N}\Big\{\tilde{\omega}_{1x}^T(i)(\Phi_x^T)^{i-1}\Phi_x^{j-1}\tilde{\omega}_{1x}(j)$$

$$+\tilde{\omega}_{1x}^T(i)(\Phi_x^T)^{i-1}\Phi_x^{j-1}G\tilde{d}_s + \tilde{d}_s^T G^T(\Phi_x^T)^{i-1}(\Phi_x)^{j-1}\tilde{\omega}_{1x}^T(j)\Big\}$$

$$+ 2(1-\beta)^{-2}\sum_{i=0}^{N}\lambda_{\max}(G^T(\Phi_x^T)^{i-1}\Phi_x^{i-1}G)\tilde{d}_s^2$$

$$+ 2(1-\beta)^{-2}\sum_{i=0}^{N}\sum_{j=0, j\neq i}^{N}\tilde{d}_s^T G^T(\Phi_x^T)^{i-1}\Phi_x^{j-1}G\tilde{d}_s$$

$$\tilde{\omega}_{1x}(k) = \tilde{\omega}_x(k) - G\tilde{d}_s,\ \ \tilde{\omega}_x(k) = \tilde{B}_1 e_{wf}(k) + G\tilde{d}_s + GO(T^2)$$

$$\tilde{B}_1 = [BV\ \ B]$$

with $\forall\beta_2 > \lambda_{\min}(Q)$ and $\beta_2 > \beta_1 > 0$.

Proof. To begin with, we aim to prove the recursive feasibility of Problem 1. Assume that there exists a solution to Problem 1 at time k with the optimal input sequence $\{\bar{u}^*(k|k), \bar{u}^*(k+1|k), \ldots, \bar{u}^*(k+N-1|k)\}$. Consider the following feasible control sequence as $u_{fe}(s|k+1) = \bar{u}^*(s|k)$ for $s \in [k+1, k+N)$ and $u_{fe}(s|k+1) = K\bar{x}^*(s|k)$ for $s \in [k+N, k+N+1)$. Because $\bar{u}^*(s|k)$, $k \in [k, k+N-1]$ is the solution to Problem 1, constraint (31a) is verified for all $\bar{x}(s|k)$ for $s \in [k, k+N-1]$.

At time $k+1$, if the feasible control input $u_{fe}(s|k+1)$ is used by system (7), then constraint (31a) naturally holds for $s \in [k+1, k+N-1]$. For $s = k+N$, in view of (35a), (35b) and (38a), we have $b_x^T\bar{x}(k+N|k) \le h_x - ((1-l_x)\sqrt{1/l_x} + \sqrt{l_x})\sqrt{b_x^T\hat{\Pi}_x(k+N|k)b_x}$ such that constraint (31a) is verified for $s \in [k+1, k+N]$. Similarly, constraint (31b) is verified for all $\bar{u}(s|k)$ for $s \in [k, k+N-1]$ at time k. Furthermore, in view of (35a), (35b) and (38b), we have $b_u^T K\bar{x}(k+N|k) \le h_u - ((1-l_u)\sqrt{1/l_u} + \sqrt{l_u})\sqrt{b_u^T\hat{\Pi}_u(k+N|k)b_u}$ such that constraint (31b) is verified for $s \in [k+1, k+N]$. According to (35a) and the invariant property (36), we have $\bar{x}(k+1+N|k) = \Phi_x\bar{x}(k+N|k) \in \mathcal{X}_f$. Furthermore, it follows from (35b) and (37) that

$$\begin{aligned}
&\hat{\Pi}_x(k+N+1|k)\\
\le&(1+\tilde{\lambda}_1+\tilde{\lambda}_2+\tilde{\lambda}_3+\tilde{\lambda}_4)\Phi_x\mathbb{E}\{\mu_x\mu_x^T\}\Phi_x^T\\
&+(4+1/\tilde{\lambda}_1)\|e_{wf}(k+N-1)\|_2^2 GG^T + G\mathbb{E}\{d_s(k)d_s^T(k)\}G^T\\
&+(4+1/\tilde{\lambda}_2)\|e_{wf}(k+N-1)\|_2^2 GVV^TG^T\\
&+(4+1/\tilde{\lambda}_3)\delta_f^2 GG^T + (3+1/\tilde{\lambda}_4)\tilde{d}_s^2 GG^T = \bar{\Pi}_{xx}
\end{aligned}$$

Hence, the constraints (35a) and (35b) are both verified at time $k+1$. This completes the proof of recursive feasibility.

Secondly, the stability of system (7) is shown. We first prove that the nominal system converges to the origin. Then, we show the state of system (7) converges to an invariant set along a trajectory lying in the tube, whose center is the trajectory of the nominal system. Select the Lyapunov function for the nominal system as $V(k) = J(\bar{x}^*(k), \bar{u}^*(k))$. Then, we have

$$\begin{aligned}
&J(x_{fe}(k+1), u_{fe}(k+1)) - J(\bar{x}^*(k), \bar{u}^*(k)) \\
=&x_{fe}^T(k+N|k)\big(Q + K^T RK + \Phi_x^T P\Phi_x - P\big)x_{fe}(k+N|k) \\
&- \|\bar{x}^*(k|k)\|_Q^2 - \|\bar{u}^*(k|k)\|_R^2
\end{aligned}$$

where x_{fe} represents the feasible state. According to (40) in Lemma 2, we have

$$\begin{aligned}
&J(x_{fe}(k+1), u_{fe}(k+1)) - J(\bar{x}^*(k), \bar{u}^*(k)) \\
\leq& - \|\bar{x}^*(k|k)\|_Q^2 - \|\bar{u}^*(k|k)\|_R^2
\end{aligned}$$

According to the principle of optimality, we have $J(\bar{x}^*(k+1), \bar{u}^*(k+1)) \leq J(x_{fe}(k+1), u_{fe}(k+1))$. Therefore, we have

$$\begin{aligned}
V(k+1) - V(k) =& J(\bar{x}^*(k+1), \bar{u}^*(k+1)) - J(\bar{x}^*(k), \bar{u}^*(k)) \\
\leq& -\|\bar{x}^*(k|k)\|_Q^2 - \|\bar{u}^*(k|k)\|_R^2
\end{aligned}$$

which implies

$$\sum_{k=0}^{\infty}(\|\bar{x}^*(k|k)\|_Q^2 + \|\bar{u}^*(k|k)\|_R^2) \leq J(\bar{x}^*(0), \bar{u}^*(0)) - \lim_{k\to\infty} J(\bar{x}^*(k), \bar{u}^*(k))$$

In other words, we have $\lim_{k\to\infty}(\|\bar{x}^*(k|k)\|_Q^2 + \|\bar{u}^*(k|k)\|_R^2) = 0$, that is, the state of the nominal system asymptotically converges to the origin. Therefore, we have $\beta_1\|\bar{x}^*(k)\|_2^2 \leq J(\bar{x}^*(k), \bar{u}^*(k)) \leq \beta_2\|\bar{x}^*(k)\|_2^2$ with $\beta_2 > \beta_1 > 0$. Because $V(k+1) - V(k) \leq -\|\bar{x}^*(k|k)\|_Q^2 - \|\bar{u}^*(k|k)\|_R^2$, we have

$$V(k+1) - V(k) \leq -\|\bar{x}^*(k|k)\|_Q^2$$

which indicates $V(k+1) \leq \left(1 - \frac{\lambda_{\min}(Q)}{\beta_2}\right)V(k)$. Selecting $\beta_2 > \lambda_{\min}(Q)$, we have $V(k) \leq \left(1 - \frac{\lambda_{\min}(Q)}{\beta_2}\right)^k V(0)$. From

$$\beta_1\|\bar{x}^*(k)\|_2^2 \leq J(\bar{x}^*(k), \bar{u}^*(k)) \leq \beta_2\|\bar{x}^*(k)\|_2^2$$

we have $\|\bar{x}^*(k)\|_2^2 \leq \frac{1}{\beta_1}\left(1 - \frac{\lambda_{\min}(Q)}{\beta_2}\right)^k J(\bar{x}^*(0), \bar{u}^*(0))$. Note that the input $\bar{u}^*(k)$ generated by the SMPC scheme is also the input used in the nominal system. Therefore, we have $\bar{x}^*(k) = \bar{x}(k)$.

In what follows, we will show that the state of system (7) converges to an invariant set along a trajectory lying in the tube, whose center is the trajectory of the nominal system. From (25), we have $x(k) = \bar{x}(k) + \tilde{e}_x(k)$. It is easy to find that $\bar{x}(k)$ is center of the tube system while $\tilde{e}_x(k)$ is the 'radius' that state $x(k)$ departs from the center $\bar{x}(k)$. Next, the convergence of $x(k)$ is shown. Introduce a set $\mathbb{W}$ where $\tilde{\omega}_x(k) - G\tilde{d}_s + Gd_s(k) = \omega_1(k) \in$

$\mathbb{W}$ with $\tilde{\omega}_x(k) = \tilde{B}_1 e_{wf}(k) + G\tilde{d}_s + GO(T^2)$ and $\tilde{B}_1 = [BV \;\; B]$. Furthermore, there exists a constant $\beta = \lambda_{\max}(\Phi_x^N)$ such that $\Phi_x^N \omega_1(k) \in \beta\mathbb{W}, \;\; \forall \omega_1(k) \in \mathbb{W}$. It is easy to have

$$\begin{aligned}\tilde{e}_x(\infty) =& \tilde{e}_x(N) + \sum_{i=N+1}^{\infty} \Phi_x^{i-1}\omega_1(i)\\ =& \tilde{e}_x(N) + \Phi_x^N\omega_1(N+1) + \Phi_x \times \Phi_x^N\omega_1(N+2) + \cdots\\ \leq& \tilde{e}_x(N) + \beta\omega_1(N+1) + \beta\Phi_x\omega_1(N+2) + \cdots\\ =& \tilde{e}(N) + \beta\tilde{e}(\infty)\end{aligned}$$

which implies $\tilde{e}_x(\infty) \leq (1-\beta)^{-1}\tilde{e}_x(N)$. Subsequently, one has

$$\begin{aligned}\tilde{e}_x^T(\infty)\tilde{e}_x(\infty) \leq& (1-\beta)^{-2}\tilde{e}_x^T(N)\tilde{e}_x(N)\\ \leq& (1-\beta)^{-2}\sum_{i=0}^{N}\sum_{j=0}^{N}\left\{\omega_1^T(i)(\Phi_x^T)^{i-1}\Phi_x^{j-1}\omega_1(j)\right\}\end{aligned}$$

Denoting $\omega_{1x}(k) = \tilde{\omega}_x(k) - G\tilde{d}_s$ where $\tilde{\omega}_x(k) = \tilde{B}_1 e_{wf}(k) + G\tilde{d}_s + GO(T^2)$ and $\tilde{B}_1 = [BV \;\; B]$ yields $\omega_1(k) = \omega_{1x}(k) + Gd_s(k)$. Then, we have

$$\begin{aligned}&\mathbb{E}\{\|\tilde{e}_x(\infty)\|_2^2\} \leq (1-\beta)^{-2}\sum_{i=0}^{N}\sum_{j=0}^{N}\left\{\tilde{\omega}_{1x}^T(i)(\Phi_x^T)^{i-1}\Phi_x^{j-1}\tilde{\omega}_{1x}(j)\right.\\ &\left.+\tilde{\omega}_{1x}^T(i)(\Phi_x^T)^{i-1}\Phi_x^{j-1}G\tilde{d}_s + \tilde{d}_s^T G^T(\Phi_x^T)^{i-1}\Phi_x^{j-1}\tilde{\omega}_{1x}^T(j)\right\}\\ &+(1-\beta)^{-2}\sum_{i=0}^{N}\lambda_{\max}(G^T(\Phi_x^T)^{i-1}\Phi_x^{i-1}G)\bar{d}_s^2\\ &+(1-\beta)^{-2}\sum_{i=0}^{N}\sum_{j=0,j\neq i}^{N}\tilde{d}_s^T G^T(\Phi_x^T)^{i-1}\Phi_x^{j-1}G\tilde{d}_s\end{aligned}$$

According to (25), we have

$$\begin{aligned}x^T(k)x(k) =& \bar{x}^T(k)\bar{x}(k) + \bar{x}^T(k)\tilde{e}_x(k) + \tilde{e}_x^T(k)\bar{x}(k) + \tilde{e}^T(k)\tilde{e}(k)\\ \leq& 2\bar{x}^T(k)\bar{x}(k) + 2\tilde{e}_x^T(\infty)\tilde{e}_x(\infty)\end{aligned}$$

which implies

$$\mathbb{E}\{\|x(k)\|_2^2\} \leq 2\|\bar{x}(k)\|_2^2 + 2\mathbb{E}\{\|\tilde{e}_x(\infty)\|_2^2\} = \tilde{\zeta}_3(x(0),k) + \tilde{\theta}_3$$

Therefore, system (7) is exponentially bounded in the mean square sense. The proof is complete. □

Remark 14. In Theorem 4.3, the recursive feasibility of the SMPC scheme and the exponential boundedness in the mean square sense for system are proved. The exponential boundedness in the mean square sense contains the transient performance $\tilde{\zeta}_3(x(0),k)$ and steady performance $\tilde{\theta}_3$. As $k \to +\infty$, the transient performance $\tilde{\zeta}_3(x(0),k) \to 0$ and $\mathbb{E}\{\|x(+\infty)\|_2^2\} \leq \tilde{\theta}_3$, which suggests that $\mathbb{E}\{\|x(+\infty)\|_2^2\}$ converges to a bounded region.

Remark 15. In this paper, we prove the NSS in probability of the CDO and the mean square stability of the closed-loop system separately due mainly to the separation principle. The proof of the separation principle can be easily got and it is omitted here.

Remark 16. In this paper, we propose the C-SMPC method to deal with modeled/unmodeled disturbances and state/input constraints for control systems. Though composite-disturbance-observer based controllers have been widely studied [40–43], our paper has the following contributions: 1) the proposed C-SMPC can not only deal with multi-source disturbances, but also state and input constraints, which is meaningful in practice; 2) the NSS in probability is established for the proposed CDO; 3) sufficient conditions are provided to guarantee the recursive feasibility and the mean square stability of the closed-loop system under C-SMPC.

Remark 17. Though the results only give the case of the linear case, the results are still applicable to some nonlinear cases. For example, when the nonlinear part of the nonlinear system is differentiable and its derivative is bounded, the results of this paper are applicable. In this case, the nonlinear part can be extended as a new state such that the whole nonlinear system can be translated as a linear system. Then, we can design an extended state observer (ESO) to get the estimate of this linear system. By this step, we can use the similar procedure of this paper to deal with the obtained linear system and obtain similar results of this paper.

5. Numerical example

In this section, a numerical example on the attitude control of the spacecraft is provided to validate the proposed method [39]. The inertia matrix and the sampling period are selected as $J_{s/c} = \mathrm{diag}\{17, 12, 9\}$ and $T = 0.1$s, respectively. The matrices of the exogenous system are given by

$$\tilde{W} = \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9.7 & 0 \\ 0 & 0 & 9.8 \end{bmatrix}, \ \tilde{H} = \begin{bmatrix} 100 & 0 & 0 & 0 & 0 & 0 \\ 0 & 100 & 0 & 0 & 0 & 0 \\ 0 & 0 & 100 & 0 & 0 & 0 \end{bmatrix}$$

$$\tilde{V} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The distribution function for each component $d_{s,i}$ is obtained by truncating a normal distribution whose mean is $\tilde{d}_{s,i} = 0$ and variance is $\breve{d}_{s,i} = 0.25$. Set $\bar{d}_{s,i} = 0.1$, $\bar{d}_b = 0.1$, $\bar{d}_f = 2$, and $\bar{w} = 2.5$. The parameters of the probabilistic constraints are given by $b_x = 0.01 \times [1\ 1\ 1\ 1\ 1\ 1]^T$, $b_u = 0.01 \times [1\ 1\ 1]^T$, $h_x = 10$, $h_u = 10$, $l_x = 0.5$ and $l_u = 0.5$. By solving (11), the parameters of the CDO are given as

$$L_w = \begin{bmatrix} 0 & 0 & 0 & 4.3680 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4.3680 & 0 \\ 0 & 0 & 0 & 0 & 0 & 4.3680 \end{bmatrix}$$

$$L_f = \begin{bmatrix} 0 & 0 & 0 & 5.1458 & 0 & 0 \\ 0 & 0 & 0 & 0 & 5.0748 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5.0303 \end{bmatrix}$$

By solving (40), the controller parameter is obtained as $K = [-0.1379\mathbf{I}_{3\times3} \ \ -0.7787\mathbf{I}_{3\times3}]$. Furthermore, according to Lemma 2, the terminal region is obtained as $\mathcal{X}_f = \{x(k) : x(k)^T P x(k) \le \xi_x^2\}$ with $\xi_x = 0.0092$. The rest parameters are given as $\delta_f = 0.1$, $\lambda_1 = 0.5$, $\tilde{\lambda}_1 = \tilde{\lambda}_2 = \tilde{\lambda}_3 = \tilde{\lambda}_4 = 0.0034$. Select the initial states as $[1\ 5\ \ -2\ 0.5\ 0.1\ \ -0.3]^T$.

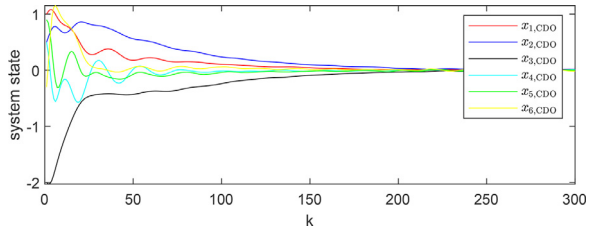

Fig. 1. The state $x_{\mathrm{CDO}}(k)$ of system (7) with the CDO.

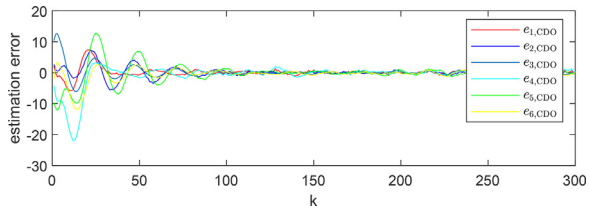

Fig. 2. The relative error $e_{\mathrm{CDO}}(k)$ of system (7) with the CDO.

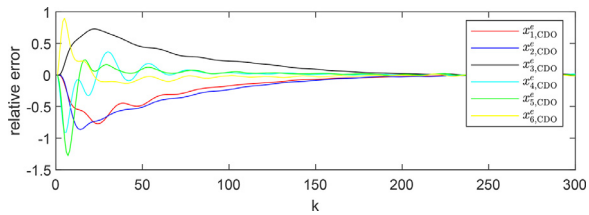

Fig. 3. The relative error $x^e_{\mathrm{CDO}}(k)$ of system (7) with the CDO.

The measurement noises are all considered as white Guassian noise with standard deviation 0.01 deg/s. Furthermore, the following comparative experiments are carried out: 1) system (7) with CDO; 2) system (7) with the DO; 3) system (7) with the ESO.

Firstly, results of system (7) using the proposed CDO are shown in Figs. 1–3. In the figures, $x_{i,\mathrm{CDO}}(k)$ $(i = 1, 2, \cdots, 6)$ represents the i-th component of $x_{\mathrm{CDO}}(k)$ which is the state of the spacecraft system using the CDO. Similarly, $e_{\mathrm{CDO}}(k)$ is the estimation error of the CDO. $e_{i,\mathrm{CDO}}(k)$ is the i-th component of $e_{\mathrm{CDO}}(k)$. $x^e_{\mathrm{CDO}}(k)$ is the error between the actual system and the nominal system of the spacecraft system using the CDO. $x^e_{i,\mathrm{CDO}}(k)$ is the i-th component of $x^e_{\mathrm{CDO}}(k)$ with $x^e_{i,\mathrm{CDO}}(k) = x_{i,\mathrm{CDO}}(k) - \bar{x}_i(k)$ where $\bar{x}_i(k)$ $(i = 1, 2, \cdots, 6)$ is the component of the nominal state $\bar{x}(k)$. Note that $e_{i,\mathrm{CDO}}(k)$ can be used to verify the effectiveness of the CDO. From Figs. 1–3, it is easy to find that the state $x_{\mathrm{CDO}}(k)$ and $x^e_{\mathrm{CDO}}(k)$ converge smoothly and rapidly.

Secondly, results of system (7) using the DO are shown in Figs. 4–6. $x_{\mathrm{DO}}(k)$, $x_{i,\mathrm{DO}}(k)$, $e_{\mathrm{DO}}(k)$, $e_{i,\mathrm{DO}}(k)$, $x^e_{\mathrm{DO}}(k)$ and $x^e_{i,\mathrm{DO}}(k)$ have the similar meaning as those in the results of the system using the CDO and hence are omitted. Furthermore, the DO is given as

$$\begin{cases} \hat{w}(k) = & v_w(k) + L_w x_{\mathrm{DO}}(k) \\ v_w(k+1) = & (W - L_w G V)(v_w(k) + L_w x_{\mathrm{DO}}(k)) - L_w(A x_{\mathrm{DO}}(k) + B u_{\mathrm{DO}}(k)) \\ u_{\mathrm{DO}}(k) = & \bar{u}(k) + K(x_{\mathrm{DO}}(k) - \bar{x}(k)) - \hat{d}_o(k) \end{cases}$$

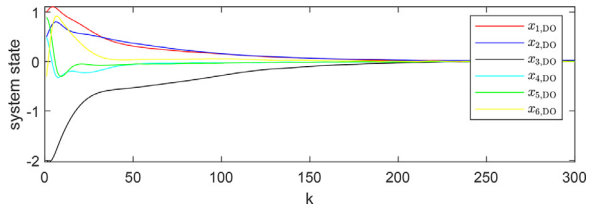

Fig. 4. The state $x_{\mathrm{DO}}(k)$ of system (7) with the DO.

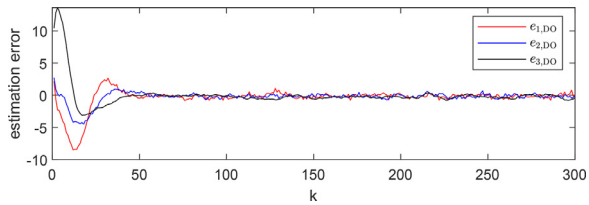

Fig. 5. The estimation error $e_{\mathrm{DO}}(k)$ of system (7) with the DO.

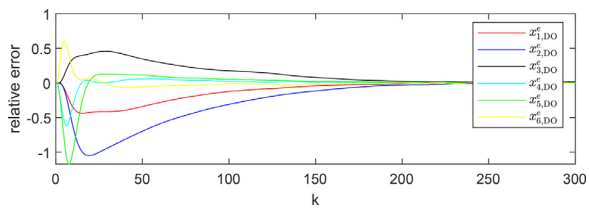

Fig. 6. The relative error $x^e_{\mathrm{DO}}(k)$ of system (7) with the DO.

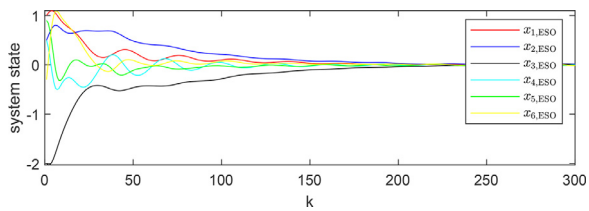

Fig. 7. The state $x_{\mathrm{ESO}}(k)$ of system (7) with the ESO.

Finally, results of system (7) using the ESO are shown in Figs. 7 and 8. $x_{\mathrm{ESO}}(k)$, $x_{i,\mathrm{ESO}}(k)$, $e_{\mathrm{ESO}}(k)$, $e_{i,\mathrm{ESO}}(k)$, $x^e_{\mathrm{ESO}}(k)$ and $x^e_{i,\mathrm{ESO}}(k)$ have the similar meaning as those in the results of the system using the CDO and hence are omitted. The ESO is given as

$$\begin{cases} \hat{d}_f(k) = & v_f(k) + L_f x(k) \\ v_f(k+1) = & (\mathbf{I}_{3\times 3} - L_f G)(v_f(k) + L_f x_{\mathrm{ESO}}(k)) - L_f(Ax_{\mathrm{ESO}}(k) + Bu_{\mathrm{ESO}}(k)) \\ u_{\mathrm{ESO}}(k) = & \bar{u}(k) + K(x_{\mathrm{ESO}}(k) - \bar{x}(k)) - \hat{d}_f(k) \end{cases}$$

In the following, quantitative evaluations for different controllers are shown in Table 1.

In Table 1, C-SMPC is the control scheme proposed in this paper. DOBC is the disturbance-observer-based control scheme using the DO. ADRC is the active disturbance rejection control scheme using the ESO. $\mathbb{E}\{e(k)\}$ represents the mean estimation error and $\|\mathbb{E}\{e(k)\}\|$ represents the norm of the mean estimation error. $\bar{k}$ represents the step that the state first enters the

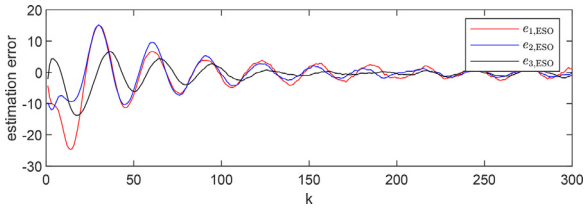

Fig. 8. The estimation error $e_{\mathrm{ESO}}(k)$ of system (7) with the ESO.

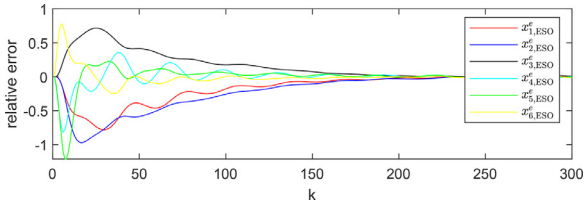

Fig. 9. The relative error $x^e_{\mathrm{ESO}}(k)$ of system (7) with the ESO.

Table 1
The quantitative evaluations for different controllers .

	C-SMPC	DOBC	ADRC
$\|\mathbb{E}\{e(k)\}\|$	0.2524 for $d_o(k)$0.9050 for $d_f(k)$	0.4854 for $d_o(k)$	1.3090 for $d_f(k)$
$\bar{k}$	90	105	108

terminal domain $\mathcal{X}_f$. Note that $\|\mathbb{E}\{e(k)\}\|$ can represent the estimation accuracy of observers. Furthermore, $\bar{k}$ can represent the convergence rate of the system. From Table 1, we can obtain that the proposed CDO improves 48.00% estimation accuracy in estimating $d_o(k)$ comparing with the DO. Besides, the proposed CDO improves 30.86% estimation accuracy in estimating $d_f(k)$ comparing with the ESO.

From the above examples, we can find that the results of the system using the CDO are better than the results using the DO/ESO, which verifies the effectiveness of the proposed method.

6. Conclusion

In this paper, the attitude control problem has been addressed for the spacecraft system in the simultaneous presence of the state/control constraint, modeled disturbance, unmodeled disturbance, and stochastic noise. To compensate for the multi-source disturbances, the CDO has been put forward and designed to provide an estimate for both modeled and unmodeled disturbances in an online manner. Then, based on the estimates provided by the CDO, the C-SMPC scheme has been designed which is capable of handling the multi-source disturbances and state/iput constraints simultaneously. Furthermore, the recursive feasibility of the C-SMPC method has been guaranteed by reformulating the state and input constraints, and the sufficient conditions have been provided to guarantee the stability of the closed-loop system. Finally, a numerical simulation on the attitude control of the spacecraft has been conducted to verify the effectiveness of the proposed method. One of the challenging works for further research is how to design a disturbance observer to obtain the estimate of $d_s(k)$ by using its information.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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